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Do Animals Lack Minds? A Critical Evaluation on Descartes' Claim

As the evolutionary outlier in cognitive complexity, humans have long claimed superiority over the rest of the animal kingdom. Although we are not the strongest or fastest species, our intelligence, creativity, and inventions of new technologies have made us the dominant species of the Earth. While other animals possess brains, they pale in comparison to humans—we can perform mathematical calculations, reflect on the past, dream and imagine different realities, etc. In his *Discourse on the Method* published in 1641 and letter to Henry More in 1649, Descartes claimed that animals lack minds. His assertion rested on two arguments: animals lack communication and the ability to solve new problems. Since then, new behavioral psychology research has emerged showing advanced cognitive abilities in various animal species, extending beyond the basic survival instincts assumed as the ceiling for animals. The question of whether non-human species lack minds has greater exigence with the rise of artificial intelligence technologies—is the human brain still paramount? In this paper, I will argue that animals do have minds, and we cannot easily accept Descartes' claim due to evidence suggesting consciousness, use of sophisticated language, and problem-solving skills present throughout the animal kingdom.

Although Descartes agrees that animals can display traits of intelligence, he clarifies that the concept of the mind means more than just processing and reacting to sensory inputs: it

requires consciousness, which will be defined and explored in the next paragraph. I will begin by explaining Descartes' justification of his claim that animals do not have minds by illustrating his two arguments. Descartes' first argument is that animals cannot form sentences and higher-level communication beyond just noises, and this absence of speech in animals "could only be explained in terms of animals lacking thought" (*Minds, Animal* | *Internet Encyclopedia of Philosophy*, n.d.). This strongly contrasts humans who have the capability of describing and consciously sharing subjective thoughts and reasoning. Descartes follows this rationale with the second argument, which states that humans are more adaptable for solving novel problems and animals do not have the same capacity for adapting flexibly to new situations.

Essential to Descartes' claim is the clear distinction between the mind and the body, and here I will introduce the concept of consciousness. Having studied anatomy, Descartes defined the body as purely a machine that, specifically in humans, can operate under the guidance and internal direction of the mind. In the phrase "animals lack minds," Descartes is not talking about intelligence or physical brainpower. He claims that the power of thought, the key separation between animals and humans, is not defined by intelligence but rather consciousness. The Stanford Encyclopedia of Philosophy describes two types of consciousness. The first is access consciousness, which describes the "availability of a mental representation for use in reasoning and the rational control of action" (*Animal Consciousness* (*Stanford Encyclopedia of Philosophy*), n.d.). The second sense, phenomenal consciousness, describes more about sentience and the qualitative side of consciousness. The text describes that when determining animal consciousness, it makes more sense to focus on phenomenal consciousness as "it is the sense of consciousness that matters most: the sense most relevant for ethics and welfare, the one that is most puzzling and mysterious, and the one that the emerging science of animal

consciousness should aspire to investigate” (*Animal Consciousness (Stanford Encyclopedia of Philosophy)*, n.d.). While Descartes' claim does not reference the neurobiologic structures of human brains compared to those of animals, research into animal brain structures is highly relevant to the argument to assess whether animals possess the capability for higher-level abstract thinking and consciousness.

Leading neuroscientists have demonstrated research indicating the existence of consciousness in non-human species. Neurobiologists explain that consciousness arises from complex interactions between the neocortex and sub-cortical regions like cortical columns. In her science book *The Human Advantage: A New Understanding of How Our Brain Became Remarkable*, prominent neuroscientist and Associate Professor at Vanderbilt University, Dr. Suzana Herculano-Houzel shares that within the neocortex, humans have 16 billion cortical columns compared to [nine] billion in gorillas and orangutans and around [seven] billion in chimpanzees (Herculano-Houzel, 2016). In 2012, the Cambridge Declaration on Consciousness wrote that birds “in their behavior, neurophysiology, and neuroanatomy [offer] a striking case of parallel evolution of consciousness” (Low, 2012). This document referred to new discoveries in neurobiologic substrates indicating both sentience (phenomenal consciousness) and cognitive function (access consciousness), and research is described providing evidence of “near human-like levels of consciousness” observed in African grey parrots (Low, 2012). The following paragraph will discuss the fascinating case study of the remarkable capabilities of one African grey parrot named Alex.

The American Psychological Association describes the exceptional intellectual competence of Alex from Dr. Pepperberg’s revolutionary longitudinal study. Even though parrots can mimic human language, this trait is dismissable as Descartes would claim this simply merely

meaningless repetition. Yet, the results from Alex's study extend beyond Descartes' initial impressions on the capabilities of animals. For three decades, Dr. Irene Pepperberg studied Alex and demonstrated this bird's ability to learn and understand "number, color, material, and shape levels to objects" (American Psychological Association, 2007). Upon hearing this exciting story, the public sensationalized Alex, sharing that he had "cognitive and linguistic abilities on par with" the "average four-year-old" (*Alex the African Grey Parrot - One of the Most Famous Birds*, n.d.). Dr. Pepperberg does share that any potential conversation she could have with Alex is nowhere near the advanced dialogue of humans, but this study effectively demonstrates the significant cognitive potential of parrots and other birds with similar avian brain structure.

In his *Discourse on the Method*, Descartes argues that animals cannot *talk*—although they can make noises, they are unable to form sentences, which are the smallest unit of language that can express complete thoughts. Even though there are humans who are incapable of speech, they are still able to communicate in advanced language. Since Descartes' time, we have identified signals of complex animal communication, from dolphins using structured communication and signature whistle patterns to rapidly transmit high amounts of information, to the dance of bees that communicate precise movements to convey the location and distance of food sources. The latter of this was demonstrated through a study from Virginia Tech's Department of Entomology, showing that the unique dance styles of each honeybee enhances their communication of the location of food sources (Mason, 2025). However, we have not found conclusive proof of animal communication that expands upon topics of survival, food or mating, while human language enables the communication of almost infinite concepts. Additionally, we have not identified complex hierarchical grammatical organization in animal language. Ultimately, it seems that animals are capable of some level of complex and interpretable

language, albeit nowhere near the human lexicon as far as we know, which somewhat corroborates the earlier refutations of Descartes' claim.

While it is challenging to prove whether animals can form sentences, proving problem-solving skills in animals is less daunting. Research in the past 200 years has demonstrated that many animals can solve new, unfamiliar problems by adapting their behavior, using tools, and learning from experience or observation. In 2024, Dr. Lauren Stanton published a study in the Royal Society Publishing that tests raccoons' problem solving skills. The results of the study show that the successful raccoons were able to try different approaches to tackling obstacles, which indicates flexibility. 25% of the raccoons were able to open at least three doors over the 3-month study period, demonstrating flexibility and analytical skills (*Wild Raccoons Demonstrate Flexibility and Individuality in Innovative Problem-Solving | Proceedings B | The Royal Society*, n.d.). However, even this evidence is not concrete as it can be argued that problem-solving is an evolutionary skill developed in order to avoid predators and access food and does not indicate the presence of a mind.

Now that I have presented evidence suggesting that animals possess minds, I will provide several arguments supporting the lack of minds in animals. The Stanford Encyclopedia of Philosophy shares two distinctions for arguments against animal consciousness. The similarity argument aligns with American philosopher Carruther's claim that "all animal behavior can be assimilated to the non-conscious activities of humans" (*Animal Consciousness (Stanford Encyclopedia of Philosophy)*, n.d.). The idea is that these behaviors are acted upon impulse and not powered by conscious thought. Another study describes that animals' mechanism of learning does not involve consciousness in analysis of activity in the spinal cord (*Animal Consciousness (Stanford Encyclopedia of Philosophy)*, n.d.). However, in American psychologist Gordon

Gallup's experiment with self-recognition using a mirror, he states that the behavior of chimpanzees suggests a self-awareness and consciousness in the species. Dissimilarity arguments, on the contrary, emphasize the differences between humans and animals to justify that animals lack consciousness and abstract thinking. This Encyclopedia provides an example of this reasoning by considering the earthworm. The argument follows that because the earthworm's capacities are so distinct from humans, it is illogical to claim that the earthworm species are sentient. This is used to prove that therefore animals with significantly limited abilities are simply incomparable to humans and we cannot extrapolate human consciousness to these species (*Animal Consciousness (Stanford Encyclopedia of Philosophy)*, n.d.). Nonetheless, this premise is undermined by our inability to quantify the dissimilarity between animals and humans.

The argument of whether or not animals have "minds" is a complex discussion that requires clearly defining relevant terms and providing deep reasoning from both sides. Scientific evidence aiming to prove one side of the argument can be interpreted in distinct perspectives, adding to the difficulty of forming a consensus. In order to arrive at the conclusion that animals possess minds and that Descartes' claim cannot easily be accepted in today's world, I established a clear and scientific definition of the "mind" he refers to and presented established arguments that suggest the contrary of the claim while anticipating potential sources of criticism. Then, I presented formal, developed arguments against the claim and disputed them. Overall, I believe that animals do have minds and possess consciousness and can perform high-level thinking beyond what humans can measure. From Darwin's theory of evolution, all life emerges from the same source, and it seems a hasty heuristic to imagine that it is only humans who are divined with the concept of consciousness and the power of thought. After all, Dr.erculano-Houzel's statistic portrays that humans only have 1.7 times as many cortical columns compared to the

second-place animal species: “this is remarkable, but not extraordinary” (Herculano-Houzel, 2016).

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